```
# first qusetion
# Greedy Multi-Armed Bandit
# Arithmetic Mean Method

# Take number of machines as input
n = int(input("Enter number of machines: "))

machines = {}
q_values = {}

# Take rewards for each machine
for i in range(n):
    machine = input(f"Enter machine name {i + 1}: ")
    rewards = list(map(float, input(f"Enter rewards for Machine {machine}: ").split()))

    machines[machine] = rewards

    # Calculate arithmetic mean
    q_values[machine] = sum(rewards) / len(rewards)

# Display Q values
print("\n--- Machine Q Values ---")

for machine in machines:
    print(f"Machine {machine}")
    print(f"Rewards: {machines[machine]}")
    print(f"Q({machine}) = {q_values[machine]:.2f}")
    print()

# Greedy selection
selected_machine = max(q_values, key=q_values.get)

print("Greedy Selected Machine:", selected_machine)
```

```
Enter number of machines: 3
Enter machine name 1: A
Enter rewards for Machine A:  3   2   4
Enter machine name 2: B
Enter rewards for Machine B: 6   7
Enter machine name 3: C
Enter rewards for Machine C: 5    4     6

--- Machine Q Values ---
Machine A
Rewards: [3.0, 2.0, 4.0]
Q(A) = 3.00

Machine B
Rewards: [6.0, 7.0]
Q(B) = 6.50

Machine C
Rewards: [5.0, 4.0, 6.0]
Q(C) = 5.00

Greedy Selected Machine: B
```

```
# qusetion 2
# Greedy Multi-Armed Bandit
# Incremental Q-Value Update Method

Q = {'A': 0, 'B': 0, 'C': 0}
N = {'A': 0, 'B': 0, 'C': 0}

rewards = {
    'A': [3, 2, 4],
    'B': [6, 7],
    'C': [5, 4, 6]
}

# Process each machine and its rewards
for machine in rewards:
    for reward in rewards[machine]:

        # Increment count
        N[machine] += 1

        # Incremental update
        Q[machine] = Q[machine] + (reward - Q[machine]) / N[machine]

        # Find machine with highest Q value
        best_machine = max(Q, key=Q.get)
```

```
        print("Machine:", machine)
        print("Reward:", reward)
        print("N:", N[machine])
        print("Q values:", Q)
        print("Best Machine:", best_machine)
        print()

print("Final Q Values:")
print(Q)

print("Best Machine:", max(Q, key=Q.get))
```

```
Machine: A
Reward: 3
N: 1
Q values: {'A': 3.0, 'B': 0, 'C': 0}
Best Machine: A

Machine: A
Reward: 2
N: 2
Q values: {'A': 2.5, 'B': 0, 'C': 0}
Best Machine: A

Machine: A
Reward: 4
N: 3
Q values: {'A': 3.0, 'B': 0, 'C': 0}
Best Machine: A

Machine: B
Reward: 6
N: 1
Q values: {'A': 3.0, 'B': 6.0, 'C': 0}
Best Machine: B

Machine: B
Reward: 7
N: 2
Q values: {'A': 3.0, 'B': 6.5, 'C': 0}
Best Machine: B

Machine: C
Reward: 5
N: 1
Q values: {'A': 3.0, 'B': 6.5, 'C': 5.0}
Best Machine: B

Machine: C
Reward: 4
N: 2
Q values: {'A': 3.0, 'B': 6.5, 'C': 4.5}
Best Machine: B

Machine: C
Reward: 6
N: 3
Q values: {'A': 3.0, 'B': 6.5, 'C': 5.0}
Best Machine: B

Final Q Values:
{'A': 3.0, 'B': 6.5, 'C': 5.0}
Best Machine: B
```